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query	BI / read / 16																				
title	Fake news detection																				
pattern	For \$tagX/\$dayX in [tagA/dateA, tagB/dateB], compute scoreX = count(messageX) 1. Create an induced subgraph of Persons who created a Message with Tag \$tagX on \$dateX tag: Tag name = \$tagX hasTag Message day(creationDate) = \$dateX hasCreator person: Person 2. In the subgraph, count the Messages (using the same conditions) from People with ≤ \$maxKnowsLimit friends tag: Tag name = \$tagX hasTag messageX: Message day(creationDate) = \$dateX hasCreator person: Person count ≤ \$maxKnowsLimit «opt» knows Person																				
description	Given two Tag/date pairs (\$tagA/\$dateA and \$tagB/\$dateB), for each pair \$tagX/\$dateX: • Create an induced subgraph between Persons where for each pair of Persons person1/person2, both have created a Message on the day of \$dateX with Tag \$tagX. • In the induced subgraph, only keep pairs of Persons who have at most maxKnowsLimit friends (in the induced subgraph). • For these Persons, count the number of Messages created on \$dateX with Tag \$tagX. Return Persons who had at least one Messages for both \$tagA/\$dateA and \$tagB/\$dateB ranked by their total number of Messages (descending).																				
params	<table><tr><td>1</td><td>\$tagA</td><td>Long String</td><td>(a) \$tagA/\$dateA, \$tagB/\$dateB are both selected to be a flashmob Tag/date combination (b) \$tagA/\$dateA, \$tagB/\$dateB are both selected to be a non-flashmob Tag/date combination</td></tr><tr><td>2</td><td>\$dateA</td><td>Date</td><td></td></tr><tr><td>3</td><td>\$tagB</td><td>Long String</td><td></td></tr><tr><td>4</td><td>\$dateB</td><td>Date</td><td></td></tr><tr><td>5</td><td>\$maxKnowsLimit</td><td>32-bit Integer</td><td>Selected between 3 and 6</td></tr></table>	1	\$tagA	Long String	(a) \$tagA/\$dateA, \$tagB/\$dateB are both selected to be a flashmob Tag/date combination (b) \$tagA/\$dateA, \$tagB/\$dateB are both selected to be a non-flashmob Tag/date combination	2	\$dateA	Date		3	\$tagB	Long String		4	\$dateB	Date		5	\$maxKnowsLimit	32-bit Integer	Selected between 3 and 6
1	\$tagA	Long String	(a) \$tagA/\$dateA, \$tagB/\$dateB are both selected to be a flashmob Tag/date combination (b) \$tagA/\$dateA, \$tagB/\$dateB are both selected to be a non-flashmob Tag/date combination																		
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3	\$tagB	Long String																			
4	\$dateB	Date																			
5	\$maxKnowsLimit	32-bit Integer	Selected between 3 and 6																		
result	<table><tr><td>1</td><td>person.id</td><td>ID</td><td>R</td><td></td></tr><tr><td>2</td><td>messageCountA</td><td>32-bit Integer</td><td>A</td><td>Message count for \$tagA/\$dateA</td></tr><tr><td>3</td><td>messageCountB</td><td>32-bit Integer</td><td>A</td><td>Message count for \$tagB/\$dateB</td></tr></table>	1	person.id	ID	R		2	messageCountA	32-bit Integer	A	Message count for \$tagA/\$dateA	3	messageCountB	32-bit Integer	A	Message count for \$tagB/\$dateB					
1	person.id	ID	R																		
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3	messageCountB	32-bit Integer	A	Message count for \$tagB/\$dateB																	
sort	<table><tr><td>1</td><td>messageCountA + messageCountB</td><td>↓</td><td></td></tr><tr><td>2</td><td>person.id</td><td>↑</td><td></td></tr></table>	1	messageCountA + messageCountB	↓		2	person.id	↑													
1	messageCountA + messageCountB	↓																			
2	person.id	↑																			
limit	20																				
CPs	5.3, 8.4, 8.5																				
relevance	There are two major ways to compute this query: (1) create the induced subgraph as suggested by the specification (either as a view or in materialized form), or (2) skip creating the induced subgraph and perform on-the-fly check for the number of friends (who also posted at least one Message with the given Tag on the given date). The latter approach is easier to express in systems which do not provide graph views but might result in redundant computations (the query engine might repeatedly check whether a Person has at least one Message that satisfies the conditions).																				